



Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

PhD project: New Computational Techniques for High Frequency Field Root Imagery

We invite applications for fully-funded (candidates eligible for EU and UK fees only) 4-year PhD project within the Plant-Soil Function group in the Discipline of Botany, School of Natural Sciences at Trinity College Dublin.

Project background and description

Soil respiration is the second biggest carbon flux in the terrestrial biosphere and but the atmosphere but intrinsically difficult understand and predict because it is made up of contributions from various different sources such as plants and soil microbes.

In this position were seeking a candidate interested in application of modern computational methods to a classic ecological / global change problem. You will work with data from automated 'minirhizotron systems able to make extremely high time frequency images of roots *in situ* in the field (<https://doi.org/10.1093/jxb/erac427>). The entire, multi-person project will interpret the dynamics of root properties in a field site and use these in a new way to dynamically 'partition' soil respiration between the activities of the organisms that contribute to it at scales from seasonal to sub-daily. The results will contribute to our ability to understand and predict this critical component of ecosystem function.

You will work within group spanning ecological science, engineering, and computer science methods and specifically develop aspects around advancing image interpretation of roots using machine learning, then use these to dynamically partition the measured soil respiration CO₂ flux, and apply this knowledge gained to improve process-based ecosystem models and potentially upscale to wider understanding. You will be part of a team which will also develop instruments and validate these techniques through classical plant-soil ecology. For this position, prior experience with field data, image-based root data, or ecological settings is not required but an interest in this setting is essential.

If desired, you will be able to help with fieldwork in this project in Ireland although this is not the focus of this role. The potential to use data from sites in the UK and Spain. GPU and other computing facilities are available within the wider research group and school. International collaborations, conference attendance and research trips are also expected as part of this project.

Candidate profile

The ideal candidate will:

- Hold a Bachelor or Masters degree in ecology, plant or environmental science, computer science, data science, or applied mathematics, and experience or demonstrable interest in learning the relevant cross-field techniques.

- Have well developed and demonstrable quantitative skills, including in neural network use and application, ideally in Python, R, or a similar language, and an interest in working with high-frequency and high-volume datasets gathered in 'noisy' field settings.
- Have capacity/willingness to think critically and creatively, and develop into a researcher in control of their own hypothesis-driven project
- Have interest in sustainability, global change, plant or soil science, and application of modern methods such as machine learning/computer vision and image analysis to advancing its understanding.
- Have experience in interdisciplinary projects or similar settings and a willingness to engage cross discipline.
- Meet the Trinity College Dublin [postgraduate entry requirements](#)

Funding

This is a 4-year PhD project and covers an annual tax-free stipend of €25,000, as well as project costs and EU student fees (funder constraint). EU fees apply to anyone who has worked or studied in the EU or UK for at least 3 out of the last 5 years, more details here: <https://www.tcd.ie/academicregistry/fees-and-payments/applicants/what-are-my-fees/>.

Application

Please send a CV and a 1-page personal statement detailing why you are interested in the project and names and contact details of two referees, with first review on the 6th June 2025 17:00 GMT by email to Dr. Richard Nair richard.nair@tcd.ie with the subject line '2025 Root Data PhD application'

Given sufficient suitable applications, interviews for this position will be held in June 2025. Interviews will be held online.

The personal statement should contain details of why you want to do a PhD and why you want to do this particular PhD.

We strive for a bias free recruitment process so we ask you to not send CVs that include a photo or information of a personal nature (e.g. age, marital status, nationality). Statements will be read before CVs. We encourage applications from underrepresented groups in STEM.

Please address all enquiries by email to Dr. Richard Nair, richard.nair@tcd.ie

Links

Botany at Trinity College Dublin: <https://www.tcd.ie/Botany/>

Plant Soil Function Group: <https://richardnair.github.io/>